Differentiation – Product Rule: Worked Examples

Product Rule

If $y = uv$ where u and v are functions of x, then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example

Differentiate:

$$y = 2xsinx$$

$$u = 2x; \qquad \frac{du}{dx} = 2$$

$$v = sinx; \qquad \frac{dv}{dx} = cosx$$

$$\frac{dy}{dx} = 2x(cosx) + sinx(2)$$

$$\frac{dy}{dx} = 2xcosx + 2sinx$$

This could be factorised to

$$\frac{dy}{dx} = 2(xcosx + sinx)$$